Harsh Jha

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EDUCATION

Jaypee Institute of Information Technology Noida, India
B.Tech in Computer Science and Engineering Jul 2024 - Jul 2028

TECHNICAL SKILLS

Languages: Rust, C/C++, Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
Rust & Systems: Cargo, WebAssembly, wasm-bindgen, Ratatui, WinAPI, Linux, cross-platform development
Media, ML & Tooling: FFmpeg/FFprobe, Sixel/libsixel, image-rs, Rodio, Symphonia, librosa, PyTorch, scikit-learn, Git, Docker

EXPERIENCE

AEXLRT Jun 2026 - Present
Software Engineering Intern | Flutter, Three.js, Prisma, CommonJS

- Developing end-to-end, integration, and unit tests for games across gameplay and release-critical flows.
- Fixing Flutter mobile-app bugs and contributing Blender/Three.js assets and interactions to the landing page.

Vedora May 2026 - Jul 2026
Full-Stack Development Intern | React, TypeScript, Prisma, PostgreSQL, Redis, Docker, Microsoft Azure

- Built authentication/session infrastructure, production database migrations, transaction-safe data repair tooling, and CI safeguards for a multi-tenant ed-tech platform.
- Built AI tutoring workflows with handwritten-math OCR, LaTeX rendering, input-hash caching for LLM responses, and responsive study/solver interfaces.

Virtual Waiter Platform Jan 2026 - Present
Founding Engineer | Next.js, React, Supabase, SSE, Web Push

- Building a multi-tenant real-time ordering platform with AI recommendations, tenant-aware routing, live order updates, push notifications, and load-tested browse/checkout flows.

TECHNICAL PROJECTS

WIMG - Terminal Media Viewer Oct 2025
Rust, FFmpeg/FFprobe, Sixel/libsixel, image-rs | github.com/life2harsh/wimg

- Built a Rust terminal media viewer for images, GIFs, and videos using FFmpeg and Sixel, with adaptive terminal scaling and frame-rate-aware playback up to 30 FPS.
- Developed cross-platform packaging and runtime cleanup for reliable Windows/Linux use, including bundled FFmpeg/FFprobe and a pure-Rust Sixel fallback.

Stonepen - Vector Handwriting Engine Jul 2026
Rust, WebAssembly, wasm-bindgen | github.com/life2harsh/stonepen

- Built a Rust/WebAssembly handwriting engine that captures pressure-sensitive pen, touch, and mouse input as editable vector strokes.
- Implemented stroke smoothing, geometric transforms, selection and hit-testing, image annotations, undo/redo, and JSON/SVG export through a reusable core and browser runtime.

YTuff - Terminal Music Player May 2026
Rust, Ratatui, FFmpeg, Rodio, Symphonia, WinAPI | github.com/life2harsh/ytuff

- Built a full-featured terminal music player for local audio and YouTube Music with playlists, downloads, lyrics, artwork, audio visualization, and keyboard controls.
- Designed persistent background playback with CLI control, streaming, authentication, OS media integration, and cross-platform Windows/Linux support.

EchoPath - Audio ML & Recommendation Engine May 2026
Python, librosa, PyTorch, scikit-learn, FastAPI | github.com/life2harsh/EchoPath

- Built an audio-analysis and playlist recommendation system using signal features, mel-spectrogram models, clustering, and graph-search algorithms; benchmarked search strategies across a 4,000-track dataset.

ACHIEVEMENTS

- **First Runner-Up, OSDHACK '25:** Secured second place in an international hackathon backed by FOSS United, .xyz, and CodeCrafters.
- **Google Big Code Challenge:** Ranked among the top 1,500 participants nationwide in India.
- **Competitive Programming:** Peak Codeforces rating of 1380.